

A Direct Answer to Problem 3.12 of arXiv:2509.00722

Automatic Math Discovery Run `fa.banach.001`

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Source

Richard J. Smith, *Lipschitz-free spaces and Bossard's reduction argument*, arXiv:2509.00722.

The open problem appears as Problem 3.12 on page 11 of the source PDF.

description of any of them. It could be instructive to obtain concrete examples.

I am grateful to G. Lancien for suggesting the following problem which Theorem 1.3 leaves open.

Problem 3.12. *Let M be a complete separable plu metric space. Does there exist a separable Banach space X such that $\mathcal{F}(M) \hookrightarrow X^*$?*

We finish this section with two direct applications of Lemma 3.9.

Problem Statement

The paper asks:

Let M be a complete separable plu metric space. Does there exist a separable Banach space X such that $\mathcal{F}(M) \hookrightarrow X^*$?

The notation section of the paper states that $Y \hookrightarrow Z$, for Banach spaces Y, Z , means that Y is linearly isomorphic to a subspace of Z . The proof below answers the problem in this printed sense.

Result

Theorem 1. *Let M be a separable metric space. Then there exists a separable Banach space X such that $\mathcal{F}(M)$ embeds isometrically into X^* . In fact one may take $X = \ell_1$.*

Proof. First, $\mathcal{F}(M)$ is separable whenever M is separable. Indeed, if $D \subset M$ is countable and dense and contains the base point, then the linear span of $\{\delta(x) : x \in D\}$ is dense in $\mathcal{F}(M)$.

We use the following standard fact. Let Y be a separable Banach space. There is a countable norming subset of B_{Y^*} . To see this, choose a dense sequence (y_n) in the unit sphere of Y . For each n , take $y_n^* \in S_{Y^*}$ with $y_n^*(y_n) = 1$. If $y \in S_Y$ and $\varepsilon > 0$, choose n with $\|y - y_n\| < \varepsilon$. Then

$$|y_n^*(y)| \geq |y_n^*(y_n)| - |y_n^*(y - y_n)| > 1 - \varepsilon.$$

It follows that

$$\|y\| = \sup_n |y_n^*(y)| \quad (y \in Y).$$

Define

$$T : Y \longrightarrow \ell_\infty, \quad T(y) = (y_n^*(y))_{n=1}^\infty.$$

The displayed equality says exactly that T is an isometric linear embedding. Since $\ell_\infty = (\ell_1)^*$ and ℓ_1 is separable, every separable Banach space embeds isometrically into the dual of a separable Banach space.

Apply this with $Y = \mathcal{F}(M)$. This gives an isometric embedding

$$\mathcal{F}(M) \hookrightarrow \ell_\infty = (\ell_1)^*,$$

so the required separable Banach space is $X = \ell_1$. □

Verification

The proof uses only two standard facts:

- Lipschitz-free spaces over separable metric spaces are separable.
- Every separable Banach space has a countable norming subset of its dual unit ball, hence embeds isometrically into $\ell_\infty = (\ell_1)^*$.

No computational verification is relevant here.

Limitations and Semantic Risk

This packet solves Problem 3.12 as it is literally printed, using the paper's own convention that $\hookrightarrow$ denotes linear isomorphism to a subspace.

The result does not address stronger variants that the author may have intended but did not state, for example:

- requiring the dual space X^* itself to be norm-separable;
- requiring $\mathcal{F}(M)$ to be weak-star closed in X^* ;
- requiring a complemented embedding;
- asking whether $\mathcal{F}(M)$ is itself isomorphic to a dual space.

Those variants are substantively different from the printed problem.

Verdict

Under the exact notation and wording in arXiv:2509.00722, Problem 3.12 has an affirmative answer for every separable metric space M , not only for complete separable plu spaces.